

Queue Tracker — High Level Design (HLD)

Version: 4.8.2 | **Status:** Production-Ready | **Date:** July 2026

1. Executive Summary

Queue Tracker is a unified operational platform designed for real-time customer queue management and agent productivity tracking. It eliminates data silos by providing a single source of truth for shift rosters, ticket metrics (INC/TASK), and live agent presence.

2. System Architecture

The system follows a **Modern Full-Stack Architecture** with persistent real-time communication:

- **Frontend:** React 19 Single Page Application (SPA) styled with Tailwind CSS 4.0.
- **Backend:** Node.js REST API and Socket.io WebSocket server.
- **Database:** MongoDB Atlas (NoSQL) for flexible schema and high-availability.
- **Real-Time Layer:** Socket.io for sub-second state synchronization across all clients.

3. Technology Stack

React 19

Node.js 24

MongoDB Atlas

Socket.io

Tailwind 4.0

4. Core Functional Modules

- **Productivity Tracker:** Interactive dashboard for agents to log INC (Incidents), TASK (Service Tasks), and CALL metrics.
- **Roster Management:** Dynamic scheduling system with IST/GMT timezone normalization.
- **Queue Handler (QH) Logic:** Real-time identification of designated leads for specific shift windows.
- **Admin Control Center:** User lifecycle management, role assignment, and security oversight.

5. Security & Data Integrity

- **Multi-Factor Authentication:** Mandatory TOTP-based 2FA for all accounts.
- **Encrypted Persistence:** AES-256-GCM encryption for sensitive configuration and 2FA secrets.
- **Hierarchical RBAC:** Strict Role-Based Access Control (Admin, Lead, Agent).
- **Real-Time Auditing:** Persistent logging of all UI and API mutations.

6. High-Level Data Flow

1. User authenticates via JWT-secured REST endpoint.
2. Client establishes a WebSocket connection for real-time events.
3. State changes (e.g., +1 Incident) are emitted to the server.
4. Server validates, persists to MongoDB, and broadcasts the diff to all active users.

5. React components re-render instantly to reflect the global state.